PS200C: Instrumental Variables

Instrumental Variables: Modern Perspective with Heterogeneous Treatment Effects

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The Intuition for (Modern) IV

The key intuition for most causal IV work can be thought about in terms of a *randomized encouragement* design.

Suppose you want to test an important hypothesis: in humans, does pizza consumption improve happiness?

- Can we just look at a sample and use $\mathbb{E}[happy|pizza] - \mathbb{E}[happy|no\ pizza]$?
- Turns out your IRB won't let you randomize, because denying pizza for the controls would be a human rights violation
- Instead, suppose we randomly assign half the people in our sample to receive gift cards for pizza.
- Assume we get to observe which ones actually get a pizza in the following week, then measure happiness for everybody in our sample.

The Intuition for (Modern) IV

Good news: we have an experiment w.r.t. our gift card (aka "encouragement", "instrument")

- Is the following identifiable?

$$ATE_{card} = \mathbb{E}[happy_{1i} - happy_{0i}]$$

- ATE_{card} called the *reduced form* or *intent to treat* (ITT) effect

But how does this compare to actual causal effect of *pizza* on *happiness*?

Answer: We can back-out effect of the pizza, but only for a certain group.

Backing out effect of pizza:

Intuition:

- suppose *card* impacts *happiness* only through *pizza* consumption
- our entire ITT estimate comes from just the sub-pop for whom *card* changed their *pizza* consumption.
- ATE_{card} is thus an average of: (a) the effect of *card* on *happy* among that group, and (b) a zero effect from those not driven to change their pizza consumption
- so ATE_{card} is a possibly "watered down" estimate of the effect of pizza.
- to undo this watering down, we'll see next that we want to divide the ATE_{card} by Pr(somebody only get pizza because of the card), which we can estimate too.

The Intuition for (Modern) IV

Formalizing this, let's consider four groups:

1. *Never-takers.* People with gluten allergies. Won't take the pizza whether they get card or not. What is $\mathbb{E}[Y|card] - \mathbb{E}[Y|no\ card]$ for this group?
2. *Compliers.* Those who will get the pizza if we give them the card, but not if we don't. Since they follow the encouragement, for them the encouragement *is* the treatment, so $\mathbb{E}[Y|card] - \mathbb{E}[Y|no\ card]$ *is* the ATE *for this group*.
3. *Always-takers* are those who will eat pizza regardless of getting the card. What is $\mathbb{E}[Y|card] - \mathbb{E}[Y|no\ card]$ for this group?
4. *Defiers* are those who feel like eating pizza if they don't get a card but when you give them the card, feel like eating salad. We just assume these people don't exist.

The Intuition for (Modern) IV

The ATE_{card} is averaged over all units,

$$
\begin{aligned}
ATE_{card} &= \mathbb{E}_{type}[ATE_{card}|type] \\
&= (ATE_{card}|\text{never-taker}) \cdot Pr(\text{never-taker}) \\
&+ (ATE_{card}|\text{complier}) \cdot Pr(complier) \\
&+ (ATE_{card}|\text{always-taker}) \cdot Pr(\text{always-taker}) \\
&+ (ATE_{card}|\text{defier}) \cdot Pr(\text{defier}) \\
&= 0 + (ATE_{card}|\text{complier}) \cdot Pr(complier) + 0 + 0 \\
\frac{ATE_{card}}{Pr(\text{complier})} &= (ATE_{card}|\text{complier}) = ATE_{pizza}|\text{complier}
\end{aligned}
$$

So we just need $Pr(complier)$...

The Intuition for (Modern) IV

$Pr(complier)$ = prob. take treatment when encouraged and not otherwise...

Let

- Z_i be the encouragement (card), D_i the treatment (pizza)
- D_{1i} and D_{0i} are "potential treatments", i.e. whether you take the treatment when encouraged (D_{1i}) or not encouraged (D_{0i})

So $Pr(complier) = Pr(D_{1i} = 1) - Pr(D_{0i} = 1) = \mathbb{E}[D_{1i} - D_{0i}]$

Is this identifiable?

Yes, by randomization of encouragement!

$$\begin{aligned} Pr(Complier) &= \mathbb{E}[D_{1i} - D_{0i}] \\ &= \mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0] \end{aligned}$$

Saying the Same Thing Many Ways:

We can estimate *ATE* among compliers, aka the *LATE* (local average treatment effect):

$$
\begin{aligned}
LATE &= \frac{ReducedForm}{ComplianceRatio} = \frac{ITT}{Pr(complier)} \\
&= \frac{\text{Effect of Z on Y}}{\text{Effect of Z on D}} \\
&= \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]} \\
&= \frac{cov(Z,Y)}{var(Z)} \Big/ \frac{cov(Z,D)}{var(Z)} \\
&= \frac{(Z^\top Z)^{-1}(Z^\top Y)}{(Z^\top Z)^{-1}(Z^\top D)} = (Z^\top D)^{-1}(Z^\top Y)
\end{aligned}
$$

This is the IV, 2SLS, or Wald Estimator

The Intuition for (Modern) IV

Lot's of things can go wrong:

- We had to assume no effect of Z on Y except through D.
  Exclusion restriction. (What would go wrong otherwise?)
- We had to assume Z effects D.
  Relevance. (What would go wrong otherwise?)
  - Moreover, Z cannot affect D too weakly. (Why?)
- We had to assume Z was "random"
  Ignorability. (What would go wrong otherwise?)
- We had to assume no defiers
  Monotonicity. (What would go wrong otherwise?)
- If everything goes right, is the effect you estimate a representative treatment effect?

Good habit to think through any IV problem
in this "randomized encouragement" setup

A loosely related example for intuition:

Figure: Intent to Eat

- For every Tbs of applesauce I (try to) feed Ronin, he sleeps another hour.
- But, for each Tbs I feed him, only 0.8 Tbs is actually ingested
- What is the effect of applesauce ingested on his sleep (in hours per Tbs)?
- What is the compliance rate? What is the exclusion restriction?

Two Views on Instrumental Variables

That's a suitable way to think about instruments. Now let's do it all again with more technicality. Two ways:

1. Traditional Econometric Framework
   - Constant treatment effects
   - Linearity in the case of a multivalued treatment
2. Potential Outcome Model of IV
   - Acknowledges heterogeneous treatment effects
   - This induced a focus on Local Average Treatment Effect
   - Most natural for binary treatments

Identification with Traditional Instrumental Variables

Definition

Two equations:

- First Stage: $D = \tau + \rho Z + \eta$
- Second Stage: $Y = \gamma + \alpha D + \varepsilon$

Identification Assumption

1. *Exogeneity and Exclusion:* $Cov(Z, \eta) = 0$ *and* $Cov(Z, \varepsilon) = 0$
2. *First Stage Relevance:* $\rho \neq 0$ *(typically* $F > 10$ *in first stage regression)*
3. $\alpha = Y_{1i} - Y_{0i}$ *constant for all units* i.
   (or for continuous treatments, $Y_{a',i} - Y_{a,i} = \beta(a' - a)$*.)*

Potential Outcome Model for Instrumental Variables

Definition (Instrument)

Z_i: Binary instrument for *unit i*.

$$Z_i = \begin{cases} 1 & \text{if unit } i \text{ "encouraged" to receive treatment} \\ 0 & \text{if unit } i \text{ "encouraged" to receive control} \end{cases}$$

Definition (Potential Treatments)

D_{zi} indicates *potential* treatment status given $Z_i = z$

- e.g. if $D_{1i} = 1$ encouraged to take treatment and takes treatment

Assumption

Observed treatments are realized as

$$D_i = Z_i \cdot D_{1i} + (1 - Z_i) \cdot D_{0i} \text{ so } D_i = \begin{cases} D_{1i} & \text{if } Z_i = 1 \\ D_{0i} & \text{if } Z_i = 0 \end{cases}$$

Potential Outcome Model for Instrumental Variables

Following Angrist, Imbens, and Rubin (1996), we can define:

Definition

- *Compliers*: $D_{1i} > D_{0i}$ ($D_{0i} = 0$ and $D_{1i} = 1$).
- *Always-takers*: $D_{1i} = D_{0i} = 1$.
- *Never-takers*: $D_{1i} = D_{0i} = 0$.
- *Defiers*: $D_{1i} < D_{0i}$ ($D_{0i} = 1$ and $D_{1i} = 0$).

Problem

Only one of the potential treatment indicators (D_{0i}, D_{1i}) *is observed, so we cannot identify which group any particular individual belongs to*

Potential Outcome Model for Instrumental Variables

Estimand (LATE)

$$\alpha_{LATE} = \mathbb{E}[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]$$

is the Local Average Treatment Effect among compliers.

NB: this estimand varies with the particular instrument Z.

Some special cases:

- When the treatment intake, D_i, is itself randomized,then $Z_i = D_i$ and every individual is a complier
- If one-sided noncompliance s.t. $D_{0i} = 0$:

$$\mathbb{E}[Y_{1i} | D_{1i} > D_{0i}] = \mathbb{E}[Y_{1i} | Z_i = 1, D_{1i} = 1] = \mathbb{E}[Y_{1i} | D_i = 1]$$

And

$$\mathbb{E}[Y_{0i} | D_{1i} > D_{0i}] = \mathbb{E}[Y_{0i} | D_i = 1]$$

Thus,

$$\alpha_{LATE} = \mathbb{E}[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1] = \alpha_{ATET}$$

Identification with Instrumental Variables

Identification Assumption

1. *Ignorability of the Instrument:* $(Y_{0i}, Y_{1i}, D_{0i}, D_{1i}) \perp\!\!\!\perp Z_i$
2. *First Stage:* $P(D_{1i} = 1) \neq P(D_{0i} = 1)$
3. *Monotonicity:* $D_{1i} \geq D_{0i}$ *for all i (no defiers)*

Identification Result

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0 | D_1 > D_0] &= \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]} \\ &= \frac{\textit{Intent to Treat Effect of } Z \textit{ on } Y}{\textit{First Stage Effect of } Z \textit{ on } D} \\ &= \frac{\textit{Intent to Treat Effect}}{\textit{Proportion of Compliers}}\end{aligned}$$

Identification with Instrumental Variables

Identification Assumption

1. *Ignorability of the Instrument:* $(Y_0, Y_1, D_0, D_1) \perp\!\!\!\perp Z$
2. *First Stage:* $0 < P(Z = 1) < 1$ *and* $P(D_1 = 1) \neq P(D_0 = 1)$
3. *Monotonicity:* $D_1 \geq D_0$

Proof.

$$\frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} = \frac{E[Y_0 + (Y_1 - Y_0)D_1|Z=1] - E[Y_0 + (Y_1 - Y_0)D_0|Z=0]}{E[D_1|Z=1] - E[D_0|Z=0]}$$

$$= \frac{E[Y_0 + (Y_1 - Y_0)D_1] - E[Y_0 + (Y_1 - Y_0)D_0]}{E[D_1] - E[D_0]} = \frac{E[(Y_1 - Y_0)(D_1 - D_0)]}{E[D_1 - D_0]}$$

$$= \frac{E[Y_1 - Y_0|D_1 > D_0]P(D_1 > D_0) - E[Y_1 - Y_0|D_1 < D_0]P(D_1 < D_0)}{E[D_1 - D_0]} \text{ as } (D_1 - D_0)\text{=}(1, 0, -1)$$

$$= \frac{E[Y_1 - Y_0|D_1 > D_0]P[D_1 > D_0]}{P(D_1 > D_0)} = E[Y_1 - Y_0|D_1 > D_0]$$

□

Comments on Identification Assumptions

Ignorability of the Instrument:

$$\{Y_{0i}, Y_{1i}, D_{0i}, D_{1i}\} \perp\!\!\!\perp Z_i$$

- Implies Z_i "random", so that the intent to treat effect and first stage effect are causally identified
- Gets us exclusion too: if Z_i changed outcome through channel other than treatment, it would change Y_{0i} and Y_{1i}. (See pset!)
- Remember, exclusion needed to attribute effect of Z on Y to the effect of D alone. Never fully testable!
- Random assignment *not* sufficient for exclusion. (Why not?)

First Stage/ Relevance

- Implies that the instrument Z induces variation in D
- Fully testable by regressing D on Z. Rule of thumb: get F-stat over 10 (better: use Anderson-Rubin CIs)

Monotonicity: $D_1 \geq D_0$

- Rules out defiers. Not testable but often qualitatively convincing.

Who are the Compliers? Is exclusion restriction reasonable?

Study	Outcome	Treatment	Instrument
Angrist and Evans (1998)	Earnings	More than 2 Children	Multiple Second Birth (Twins)
Angrist and Evans (1998)	Earnings	More than 2 Children	First Two Children are Same Sex
Levitt (1997)	Crime Rates	Number of Policemen	Mayoral Elections
Angrist and Krueger (1991)	Earnings	Years of Schooling	Quarter of Birth
Angrist (1990)	Earnings	Veteran Status	Vietnam Draft Lottery
Miguel, Satyanath and Sergenti (2004)	Civil War Onset	GDP per capita	Lagged Rainfall
Acemoglu, Johnson and Robinson (2001)	Economic performance	Current Institutions	Settler Mortality in Colonial Times
Cleary and Barro (2006)	Religiosity	GDP per capita	Distance from Equator

Instrumental Variable: Estimators

Estimand (LATE)

$$\mathbb{E}[Y_1 - Y_0 | D_1 > D_0] = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]}$$

Estimator (Wald Estimator)

The sample analog estimator is:

$$\left(\frac{\sum_{i=1}^{N} Y_i Z_i}{\sum_{i=1}^{N} Z_i} - \frac{\sum_{i=1}^{N} Y_i (1 - Z_i)}{\sum_{i=1}^{N} (1 - Z_i)} \right) \Bigg/ \left(\frac{\sum_{i=1}^{N} D_i Z_i}{\sum_{i=1}^{N} Z_i} - \frac{\sum_{i=1}^{N} D_i (1 - Z_i)}{\sum_{i=1}^{N} (1 - Z_i)} \right)$$

Instrumental Variable: Estimators

Estimator (Two-Stage Least Squares)

Can also implement Wald Estimator using two-stage least squares regression:

1. $D = \pi_0 + \pi Z + \eta$ *(First Stage)*
2. $Y = \beta_0 + \beta D + \varepsilon$ *(Second Stage)*

Implies

$$\begin{aligned} Y &= \beta_0 + \beta(\pi_0 + \pi Z + \eta) + \varepsilon \\ &= (\beta_0 + \beta\pi_0) + \beta\pi Z + (\beta\eta + \varepsilon) \end{aligned}$$

where $cov(\varepsilon, Z) = 0$ *due to exclusion and* $cov(\eta, Z) = 0$ *by exogeneity in first stage*

- β *gives the LATE*
- *If you just regressed* Y *on* Z*, you get effect of* Z *on* Y*, i.e.* $\beta\pi$ *(reduced form). Divide this by* π *(compliance ratio or first stage) and you get* β*.*
- *Don't actually estimate this way, you will get wrong SEs*

Instrumental Variable: Estimators

Estimator (IV Estimator)

Standard implementation is the IV estimator:

$$\hat{\beta}_{IV} = (\mathbf{Z}^\top \mathbf{D})^{-1} \mathbf{Z}^\top \mathbf{Y}$$

1. Typically estimate using *ivreg()*
2. If ID assumptions only hold after conditioning on X, you can add covariates:
   - they are added to both stages
   - $\mathbf{X} = [1; X\ \mathbf{D}]$ and $\mathbf{Z} = [1\ \mathbf{X}\ \mathbf{Z}]$ and estimated by $(\mathbf{Z}^\top \mathbf{X})^{-1} \mathbf{Z}^\top \mathbf{Y}$

Example: The Vietnam Draft Lottery (Angrist (1990))

- Effect of military service on civilian earnings
- Simple comparison between Vietnam veterans and non-veterans are likely to be a biased measure
- Angrist (1990) used draft-eligibility, determined by the Vietnam era draft lottery, as an instrument for military service in Vietnam
- Draft eligibility is random and affected the probability of enrollment
- Estimate suggest a 15% effect of veteran status on earnings in the period 1981-1984 for white veterans born in 1950-51; although imprecisely estimated.

Estimating the Size of the Complier Group

- Since we never observe both potential treatment assignments for the same unit, we cannot identify individual units as compliers!
- However, we can easily identify the proportion of compliers in the population using the first stage effect:

$$\begin{aligned} P(D_{1i} > D_{0i}) &= \mathbb{E}[D_{1i} - D_{0i}] = \mathbb{E}[D_{1i}] - \mathbb{E}[D_{0i}] \\ &= \mathbb{E}[D_{1i}|Z_i = 1] - \mathbb{E}[D_{0i}|Z_i = 0] \\ &= \mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0] \end{aligned}$$

Instrumental Variables with Covariates

- Sometimes IV identification assumptions may only hold once we condition on a set of pre-treatment characteristics X.
- In traditional econometric view of IV this poses no additional problems: treatment effects are assumed to be constant, so just include X in the 2SLS model
- But with heterogenous treatment effects, IV/2SLS does not formally identify LATE conditional on X.
- Also hard to interpret, because complier group changes depending on strata of X.
  - One approach: κ-weighting are weights s.t. any model you run gets you a result as if only estimated for compliers (Abadie, 2003)
  - When you use these weights to estimate treatment effect, you get "Local Average Response Function" (LARF)
  - But many investigators still ignore this by taking an econometric view, assuming constant treatment effects

Example: Natural Disasters and Aid

Does getting US disaster aid makes countries cooperate with the US on UN generally assembly votes

- Aid not randomly assigned in general
- Disasters aren't entirely random either, especially seasonal ones
- Non-seasonal disasters – earthquakes and volcanoes – seem totally uncorrelated with "politics"
- ...except maybe through location specific differences in probability (more soon)

So

- Encouragement: non-seasonal natural disasters that kill at least 25 people
- Treatment: getting aid from the US
- Outcome: agreement with the US at UNGA (at various times after disaster/aid)

Let's confirm:

- What is the first-stage?
- What is the reduced-form/intent-to-treat?
- Who are the compliers?

Natural Disasters and Aid with One Year Lag

```
> pd=pdata.frame(d[d$year>1963,],c("country","year"))
> rf.out=lm(daffinity~disaster_25_L1, data=pd)
> summary(rf.out)
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -0.010579   0.001353  -7.821 6.06e-15 ***
disaster_25_L1  0.008352   0.007704   1.084    0.278
---
Multiple R-squared:  0.0001759, Adjusted R-squared:  2.621e-05
F-statistic: 1.175 on 1 and 6680 DF,  p-value: 0.2784

> first.out = lm(allaid_L1~disaster_25_L1, data=pd)
> summary(first.out) #note F stat
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.127977   0.004073   31.42   <2e-16 ***
disaster_25_L1 0.308643   0.023651   13.05   <2e-16 ***
---
Multiple R-squared:  0.02317, Adjusted R-squared:  0.02303
F-statistic: 170.3 on 1 and 7181 DF,  p-value: < 2.2
```

Natural Disaster and Aid with One Year Lag

```
> iv.out=ivreg(daffinity~allaid_L1 | disaster_25_L1, data=pd)
> summary(iv.out)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.014082   0.003744  -3.761 0.000171 ***
allaid_L1    0.027133   0.025223   1.076 0.282085
---
Multiple R-Squared: -0.01529, Adjusted R-squared: -0.01544
Wald test: 1.157 on 1 and 6680 DF,  p-value: 0.2821
```

Compare to simple Wald Estimator:

```
> rf.out$coef[2]/first.out$coef[2]
disaster_25_L1
    0.02705907
```

Two Extensions

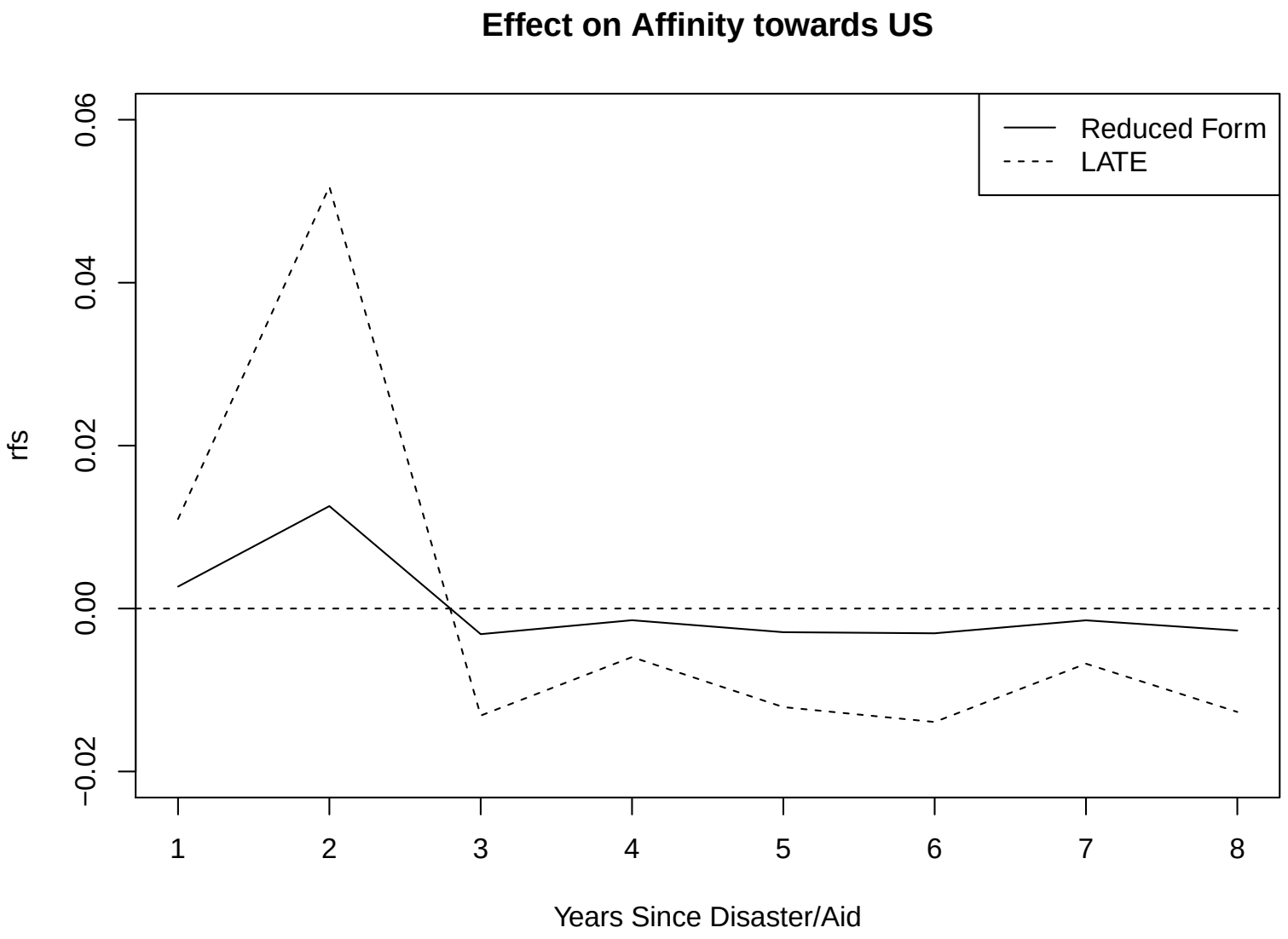
First, we probably want country FEs to strengthen the identification

```
> iv.fe.out=ivreg(daffinity~allaid_L1+country | disaster_25_L1+country, data=pd)
> coef(iv.fe.out)[2]
 allaid_L1
0.05312396
```

Second, we don't know what lag to look at so we can look at all of them. Could just run multiple models with different lags, e.g.

```
> rf0.out=lm(daffinity~disaster_25+country, data=pd)
> rf1.out=lm(daffinity~disaster_25_L1+country, data=pd)
> rf2.out=lm(daffinity~disaster_25_L2+country, data=pd)
...
> fs.out=lm(allaid~disaster_25+country, data=pd)
> fs1.out=lm(allaid_L1~disaster_25_L1+country, data=pd)
> fs2.out=lm(allaid_L2~disaster_25_L2+country, data=pd)
```

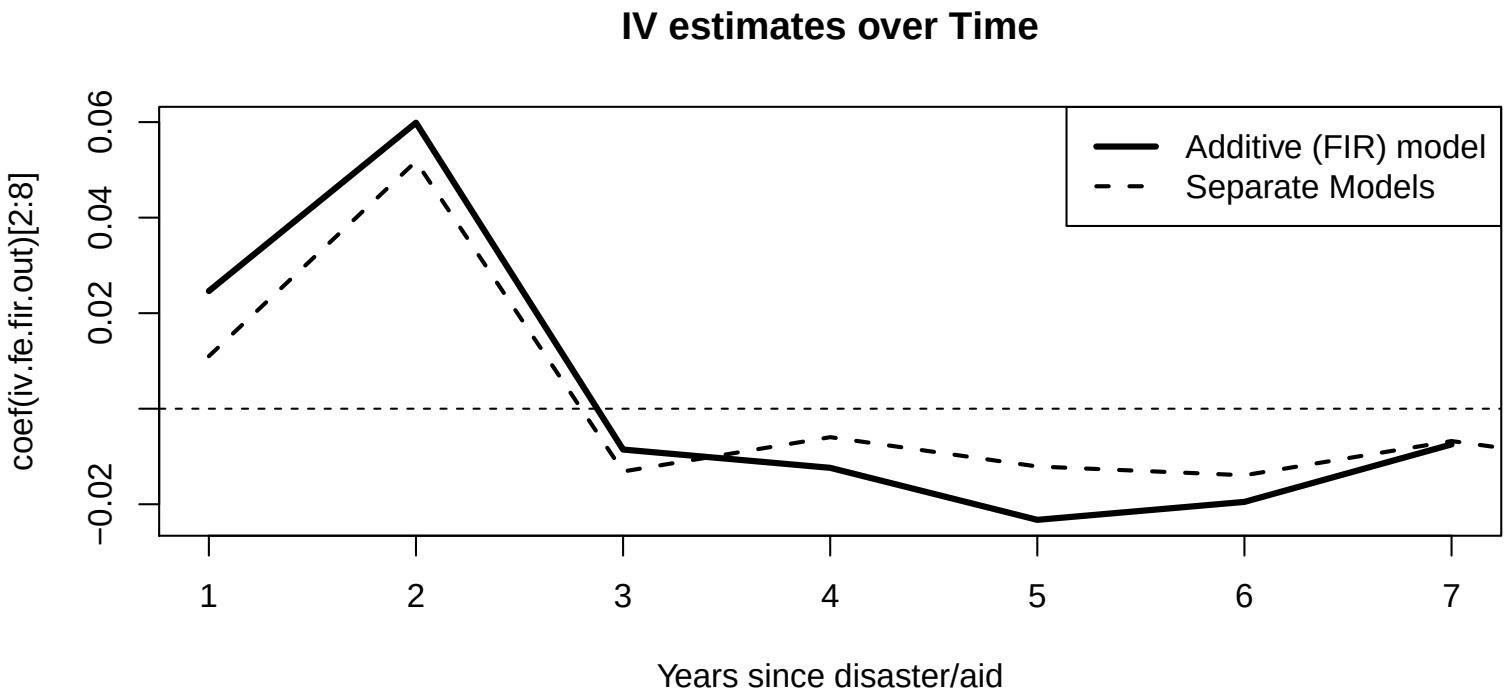
Effects Over Time



Finite-Impulse-Response, Distributed Lag, or Event Study Model

Alternatively, if disasters occur frequently enough that the effects might overlap at some time-points, we can assume additivity and do "distributed lag" or "finite impulse response" or "event-study" model. e.g.

```
iv.fe.fir.out=ivreg(daffinity~allaid+allaid_L1+allaid_L2+allaid_L3+ ...+country
    |  disaster_25+disaster_25_L1+disaster_25_L2+disaster_25_L3+...+country, data=pd)
```



Summary of Modern IV

- Instruments can give you causal identification where you don't have SOO or other options.
- The key concern is usually the exclusion restriction:
  - together with first-stage ignorability, captured by

$$\{Y_{0i}, Y_{1i}, D_{0i}, D_{1i}\} \perp\!\!\!\perp Z_i$$

  - a randomized Z_i is a good start, but not sufficient
  - plausibility tests possible, but cannot be confirmed on observables
- Generally useful to think about in terms of an "encouragement" to take the treatment and rescaling the ITT (RF) by the compliance ratio
- If you do not assume constant treatment effects, then you must recognize the estimate as a LATE among the compliers